

CALEB COOPER

Software Engineer

CONTACT

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TECHNICAL COMPETENCIES

Languages

Python, Java, JavaScript, TypeScript, C#, Rust, SQL, HTML, CSS

Technologies/Frameworks

React Native, ReactJS, Spring Boot, FastAPI, Flask, .NET, Vite, JUnit, Playwright, Docker, nginx, CI/CD Pipelines, Git, REST APIs, PostgreSQL, Redis, SQLite

PROJECTS &

EXTRA-CURRICULAR

Cartsy

Co-building a cross-platform React Native app to help students and low-income households save money on groceries, aggregating daily-updated pricing data from NZ supermarkets. Backend infrastructure is split across a self-hosted PostgreSQL and BullMQ stack for scraping and data storage, and Azure for cloud services. Features include barcode scanning, price-drop alerts, shopping list management, and price trend visualisation - currently in active TestFlight beta.

SENG402 Capstone – Meal Tracking App

Building a food logging app with an NLP pipeline combining hybrid retrieval (FAISS semantic search, BM25, reciprocal rank fusion) to match user descriptions against NZ nutritional food databases. Integrating a computer vision volume estimation module using depth models and a FastAPI inference server. Running LLMs locally via Ollama (Qwen2.5 7B) to power entity extraction.

PDF to Anki Flash Card Converter

Built a web app to convert PDF lecture notes into Anki flashcards using the OpenAI API, after finding it significantly sped up my own exam revision. Python/Flask backend containerised with Docker and served via nginx, with a clean frontend offering configurable settings including GPT model selection and card formatting options.

IMC Prosperity 2025

As a team, competed against 22,600 participants and over 12,600 teams globally in a 15-day algorithmic trading challenge. Developed and implemented both algorithmic and manual trading strategies, finishing in the top 10% overall.

EXPERIENCE

Trimble NZ – Software Developer in Test Intern

November 2025 – February 2026

- Working with Trimble Forestry, supporting the development and testing of forestry software products in a Scrum environment.
- Developed and maintained Playwright-based automated test suites for forestry software products, covering regression and functional testing across multiple release cycles.
- Worked closely with developers and QA engineers to coordinate testing efforts and improve software quality.

Allied Telesis Labs NZ – Software Engineering Summer Intern

November 2024 – February 2025

- Tasked with investigating a replacement for an outdated IxANVL automated standards-based testing suite.
- Working with a highly experienced team, exchanging ideas and gaining valuable industry insights.
- Quickly adapted to enterprise networking equipment despite no prior experience.
- Researched AI integration into the company's development process and presented key recommendations that influenced their AI adoption strategy.

Vodafone NZ – Summer OSS/BSS Software Engineering Intern

November 2022 – February 2023

- Redesigning an event monitoring dashboard using Grafana to deliver easier-to-understand status information of key services to the developers.
- Applied C# and SQL in a production Scrum environment. This gave me key beneficial insights into how teams work efficiently in industry.
- Presented final work to the team, improving my ability to communicate technical findings effectively.

SPORTING AND INTERESTS

Skiing, Football, Running, Bouldering, Social sports, Cooking, Home networking & self-hosting

REFERENCES

Available on request.

ACHIEVEMENTS

- University of Canterbury Hiranga Scholarship for Academic Excellence
- Duke of Edinburgh Bronze, Silver and Gold Awards
- Spirit of Adventure 10-Day Development Voyage

EDUCATION

University of Canterbury

February 2023 – November 2026

- Currently in my final year of study towards a Bachelor of Engineering with Honours, majoring in Software Engineering. (7.25 GPA)

Auckland Grammar School

February 2018 – November 2023

- Studied and completed Cambridge A Level History, Economics and Mathematics as well as English, Spanish, and Classical Studies at Cambridge AS Level